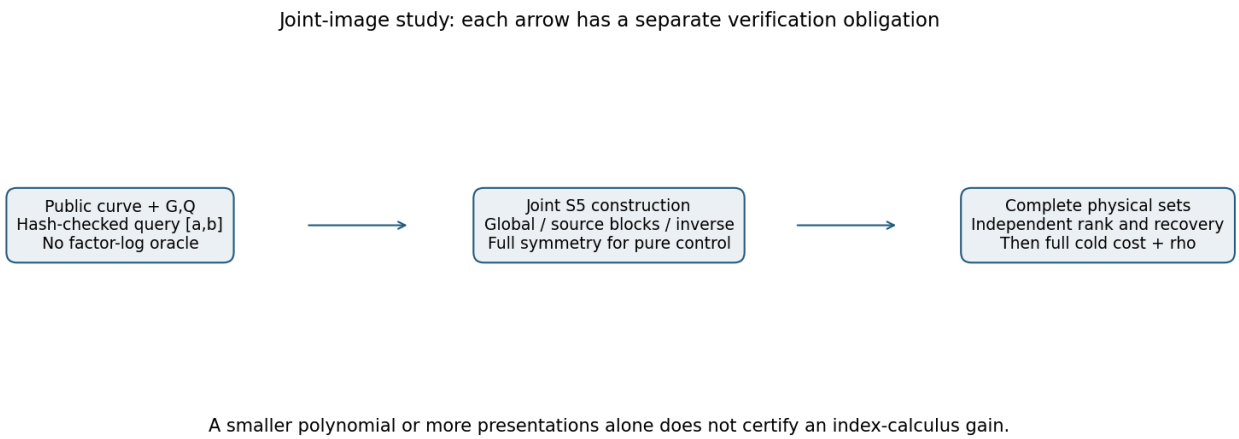
Split source-block coordinate proofs pass

Gate coordinate-lean-v4 checks the unchanged four coordinate statements in Coordinates.lean (../../proof/orbit-v2/Coordinates.lean). The metadata preserves the exact command, source hash, source snapshot, launch and finish times. This separates the coordinate obligations from the larger quartic expansion without weakening their statements.



The build and axiom audit pass all four theorems: phi and doubling cross products, and their respective specializations under U^12=T. They hold over any Lean.Grind.CommRing with the stated orbit hypothesis for the specializations. The audit log (../../results/proof-checks/coordinates-v4/audit.log) records only propext, Classical.choice and Quot.sound, with zero failing and zero trusted-native theorems.

The native IDC checks separately certify the exact integer cross products. Field division, nonzero denominators, curve-map interpretation and source partition membership remain separate bridges; native fixture checks cover the declared 21 cases. These proofs do not certify elimination-ideal equality, complete roots, independent rank or speed.

Pinned Lean 4.34.1, one worker, 8 GiB, a 900 s watchdog and the busy reservation are retained. No comparable performance values are admitted.